



COMSATS University Islamabad
Attock Campus
Department Of Computer Engineering

Complex Engineering Problem

Course Code:	Course Title:	Session:	Class:
CPE344	Digital Systems Design	Fall 2025	CE-7
Title of Complex Engineering Problem	Design HDL-based encryption/decryption unit (e.g., AES, RSA, SHA) for secure communication AES: Advanced Encryption Standard, RSA: Rivest-Shamir-Adleman, SHA: Secure Hash Algorithm Design Requirements: Students will design a mathematical model, architecture, and Verilog HDL code for cryptographic hardware engine. The engine will perform encryption and decryption of digital data using widely adopted cryptographic standards such as AES, RSA, or SHA. The project requires students to: • Understand the mathematical and algorithmic foundations of cryptography. • Design a customized architecture for an encryption standard (AES, RSA, or SHA) • Translate architecture into an optimized hardware designs in Verilog.		



COMSATS University Islamabad
Attock Campus
Department Of Computer Engineering

CEP Requirements

Statement

Preamble:

With the increasing need for **secure communication** in online banking, e-commerce, healthcare, and defense, encryption plays a crucial role in ensuring **data privacy** and **security**. As digital systems engineers, your task is to design a **cryptographic hardware engine** using **Verilog HDL**.

Your design must be able to:

1. Perform **encryption and decryption** using a standard algorithm (AES, RSA, or SHA).
2. Optimize for **speed, area, and power consumption**.



COMSATS University Islamabad
Attock Campus
Department Of Computer Engineering

WP1	Depth of Knowledge Required: In this CEP, you will study and research multiple cryptography standards. Your solution for cryptography hardware will include multiple modules. You must justify the selection and importance of your solution.
WP2	Range of Conflicting Requirements: You will trade-off among design size, energy efficiency, and level of encryption and security of your solution. For example 1. Tradeoff size and number of architecture components for parallelism Vs. energy efficiency 2. Energy efficiency Vs. processing speed 3. Energy efficiency Vs. security and encryption level Also you will choose one of the encryption standard based on your research and targeted variable (e.g. encryption level)
WP3	Depth of Analysis Required: You are expected to conduct an in-depth research and analysis and based on your analysis find: 1. Encryption standard with key length 2. Architecture i.e. to be more suitable for FPGA implementation 3. Interfaces required i.e. UART/SPI/I2C communication to demonstrate secure data transfer
WP4	Familiarities of Issues:
WP5	Extent of Applicable Codes:
WP6	Extent of Stakeholder involvement and level of conflicting requirements:
WP7	Interdependence:
EA1	Have significant consequences in range of context:
EA2	Require judgment in decision making:



COMSATS University Islamabad
Attock Campus
Department Of Computer Engineering

Deliverables

1. Technical Report

- Design Documentation: Prepare a comprehensive design document that details your system's architecture. This should include:
 1. Mathematical Models
 2. System Architecture Diagrams
 3. Analysis of different cryptography standards showcasing trade-offs in design (performance, security, energy efficiency).
 4. Hardware Diagrams that can be used as base for designing HDL codes
- System Diagram: Provide a schematic or block diagram that clearly illustrates the flow of your designed system.

2. Presentation

- Design Overview: Deliver an engaging presentation that explains your engineering decisions, the performance of your designed system, the challenges you encountered, and the innovative solutions you implemented.
- Visual Aids: Use visuals such as flowcharts, and dry-run encrypted/decrypted text etc., to clearly communicate your findings and insights.



COMSATS University Islamabad
Attock Campus
Department Of Computer Engineering

Rubrics for Assessment

Complex Problem-Solving Attributes	Indicator	WP	Level 1 Needs Work	Level 2 Developing	Level 3 Satisfactory	Level 4 Competent	Level 5 Excellent	Marks
Depth of Knowledge Required	Analyze the problem using specified knowledge profile (WKs)	WP1	Use 2 WKs but do not elaborate	Use 2 WKs with acceptable elaboration	Use 3 WKs with acceptable elaboration	Use 4 WKs with acceptable elaboration	Use more than 4 WKs with acceptable elaboration	10
	Evaluate the problems under such circumstances to provide an effective solution.	Must Have	Evaluate 1 circumstance	Evaluate 2 circumstances with acceptable justification.	Evaluate 3 circumstances with acceptable justification	Evaluate 4 circumstances with acceptable justification	Evaluate more than 4 circumstances with acceptable justification	5
Range of Conflicting requirements = wide & conflicting technical, engineering & other issues	Compare the conflicting technical, engineering and other issues to solve the problems	WP2	Provide only 1 issue	Compare 2 issues with acceptable discussion.	Compare 3 issues in accordance.	Compare 4 issues with acceptable discussion.	Compare more than 4 issues with acceptable discussion.	5
	Assess the conflicting requirements and provide a satisfactory proposal towards solving the problems.		Assess but no proposal.	Assess with 1 proposal.	Assess with 2 proposals.	Assess with 3 proposals.	Assess with more than 3 proposals.	10
Depth of analysis = no obvious solution, abstract thinking, originality	Develop the formulae/procedures to solve the problem using suitable models	WP3	Conceptualize 1 formula/procedure used	Conceptualize 1 formula/procedure used but do not elaborate on the model.	Conceptualize 1 formula/procedure used and elaborate the model.	Develop 2 formulas/procedures used and elaborate the model.	Develop 3 formulas/procedures used and elaborate the model.	10
	Justify creativity towards the achievement of the formulae/procedures.		Justify 1 creative development.	Justify the 1 creative development used but do not elaborate on the model	Justify the 1 creative development used and elaborate the model	Justify the 2 creative developments used and elaborate on the model	Justify the 3 creative developments used and elaborate on the model.	10



COMSATS University Islamabad
Attock Campus
Department Of Computer Engineering

Complex Engineering Problem Rubrics (Form)		
Project Title:	Design HDL-based encryption/decryption unit (e.g., AES, RSA, SHA) for secure communication AES: Advanced Encryption Standard, RSA: Rivest-Shamir-Adleman, SHA: Secure Hash Algorithm	
Subject	Digital Systems Design Lab	
Submitted by:		
Names		Registration #

Mapping with Graduate Attributes

Sr. #	Program learning Objective (PLO)	Course learning Outcome (CLO)	Complex Engineering Activities Attributes	Marks		
1	PLO1	CLO1	WP1			
2.	PLO2	CLO2	WP2			
3.	PLO3	CLO3	WP3			